Dave Rich

Software Engineer

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Software Engineer with seven years of experience specializing in Go, Kubernetes, containers, CI/CD, and Linux. Over fifteen years of experience using and maintaining Linux systems. Strong engineering fundamentals with AI-powered tools enable me to build, iterate, and ship high quality software faster than ever.

SKILLS

Programming: Go (Golang), TypeScript, JavaScript, Node.js, React, PHP, Python, SQL
DevOps: Kubernetes, Docker (containers), Jenkins, GitHub Actions, Argo CD, GitOps, CI, CD, Grafana, Prometheus, Kibana, Ansible, Terraform, IaC, Virtualization (VMware), AWS, Red Hat, Ubuntu, Linux, MySQL, Postgres
Project Management: Scrum, Agile, Kanban, Jira, GitHub

WORK EXPERIENCE

Professional Development November 2025 – Current

Pittsburgh, PA

- Achieved Certified Kubernetes Application Developer (CKAD) certification
- Configured lab virtual environments with Proxmox to study for Certified Kubernetes Administrator (CKA) and CKAD exams
- Streamlined software development workflows using locally hosted open-source LLMs and Claude Code
- Experimented with Rust and AI to build a desktop application which optimizes rapid context switching

Cloud Software Engineer November 2023 – October 2025

Webbula, Wexford, PA

- Created DNS parsing library for validating SPF, DKIM, and DMARC TXT records according to RFC specifications with Go
- Engineered a proof-of-concept monitoring system for scheduled detection and reporting of customer DNS configuration issues using Go, AWS SQS, ECS, and containerized Scheduler-Agent-Supervisor architecture
- Discovered novel email address verification method & architected proof-of-concept with Python scaling up to 1 million address verifications per hour
- Automated EmailHygiene product environment provisioning with Terrfarm, AWS IAM, Elastic Beanstalk, EC2, and S3 which reduced creation from 2 days to less than 30 minutes
- Strengthened 5 core company projects with Continuous Integration (CI) using GitHub Actions, Go unit tests, static analysis, and reproducible Docker builds
- Restructured form data ingestion process with Apache Airflow, Python, Amazon S3, Wordpress, and PHP
- Standardized and documented development environments which enabling rapid onboarding of new team members for organization expansion

Medical Leave May 2022 – October 2023

Pittsburgh, PA

- Returned to full capacity October 2023

Software Engineer September 2017 – April 2022

Expedient, Pittsburgh, PA

- Developed internal ticketing & infrastructure change management application using PHP, JavaScript, SQL, and Symfony framework
- Led architecture and implementation of cloud native REST API backend microservices using Go and Gin framework
- Spearheaded adoption of Go programming language for microservices and ETL workflows which increased team velocity by 20% after 3 months
- Adopted Infrastructure as Code (IaC) practices by deploying Kubernetes and MySQL on Ubuntu Linux with Terraform, Rancher, and Ansible
- Automated deployments of Kubernetes manifests & CRDs using the GitOps pattern via ArgoCD

- Optimized multi-region ETL by leveraging Go goroutines and channels which reduced runtime by 90%
- Pioneered developer workflow using Kubernetes, Skaffold, and shell scripts which led to a speed up in provisioning developer environments from 5 days to 3 hours
- Designed architecture for multi-region high availability automated failover which enabled team to exceed uptime goal of 99.99%
- Championed adoption of Kubernetes for container orchestration resulting in a reduction of virtual infrastructure by 50%
- Conducted technical interviews with candidates for DevOps and Software Engineer roles
- Resolved critical issues during 24/7 on-call rotation using Grafana, Prometheus, Elasticsearch, and Kibana monitoring and logging systems

Senior Technician April 2017 – September 2017

Expedient, Pittsburgh, PA

- Developed custom infrastructure alert processing application with Angular, TypeScript, and Node.js which reduced time to onboard new monitoring analysts by 80%
- Managed customer and internal RHEL, CentOS, and Ubuntu Linux servers by patching, troubleshooting issues, and performing online disk expansions
- Used Linux CLI (Bash) and common utilities such as awk, sed, grep, lsof, and netstat to troubleshoot issues and perform data restoration from EMC NetWorker backups
- Promoted to Senior Technician and received Q3 2017 Internal Support award from Expedient for organizational impact of custom infrastructure alert processing application

Technician March 2015 – April 2017

Expedient, Pittsburgh, PA

- Solved issues for managed backup environment of Windows and Linux file system, Microsoft Exchange, Microsoft SQL Server, and MySQL data for over 6,000 servers
- Responded to customer emergency by optimizing restoration of over 1600 production MSSQL databases with PowerShell to reduce downtime
- Enabled rapid onboarding of a new customer by converting over 11,000 DNS records to CSV with Node.js because a DNS zone transfer was not possible
- Received April 2016 Client Care Extra Effort Award from Expedient for quickly and efficiently resolving urgent customer issues by scripting solutions

Analyst August 2014 – March 2015

Expedient, Pittsburgh, PA

- Received customer support phone calls, handled ticket intake, and triaged high severity issues
- Updated virtual machine resource configurations with VMware vSphere according to customer requirements
- Earned promotion 5 months ahead of schedule for technical expertise across UNIX, Windows, backup, networking, and virtualization disciplines

CERTIFICATIONS

Certified Kubernetes Administrator (CKA)

Cloud Native Computing Foundation

Expected completion November 2026

Certified Kubernetes Application Developer (CKAD)

Cloud Native Computing Foundation

October 2026

AWS Certified Cloud Practitioner

Amazon Web Services

September 2023

Recertified September 2026